Q.1 If a vector space V has a basis of n vectors then every basis of V must consists of exactly --------

A. n vectors

B. n-1 vectors

C. n+1 vectors

D. n+2 vectors

Q.2 The determinant of identity matrix is?

A. 0

B. None of the mentioned

C. Depends on the matrix

D. 1

Q.3 If a vector space V has a basis B={b_1, b_2, ..., b_n} then any set in V, containing more than n vectors, must be ----------

A. Linearly dependent

B. Linearly Independent

Q.4 A set of two non-zero vectors are linearly independent if and only if they are ---------- to/of each other.

A. Scalar multiple

B. Equal

Q.5 The vectors spaces F(-\infty, +\infty) and C(-\infty, +\infty) are ----------

A. Infinite-dimensional

B. Finite-dimensional

Q.6 The act of changing A into P^-1 AP is called a similarity transformation

A. False

B. True

Q.7 The point of intersection of axes in three dimension coordinates system is called?

A. Ordinate

B. Collective intercept

C. Collective coordinate

D. Parallel pair order

Q.8 Rank of the matrix A=\left[\begin{gathered} 0\,\,\,0\,\,\,0\,\,\,0 \hfill \\ 4\,\,\,6\,\,\,1\,\,\,0\, \hfill \\ 1\,\,\,\,\,0\,\,\,\,\,0\,\,\,\,\,0 \hfill \\\ \end{gathered} \right]

A. 3

B. 0

C. 2

D. 1

Q.9 If~ A ~is~ an~ m \times n ~matrix ~then

A. rank A = Nul A

B. rank A + NulA = n

C. rank A = n

D. rank A = NulA + n

Q.10 The pivot columns of a matrix A form a basis for ----------

A. ColA

B. RowA

Q.1 If one of the Eigenvalues of {\left[A \right]_{n \times n}} is zero, it implies

A. The solution to [A][X]=[C] system of equations is unique

B. The determinant of [A] is zero

C. The solution to [A][X]=[0]

D. The determinant of [A] is nonzero

Q.2 If A and B are n x n matrices, then det (AB) = _____.

A. det(A).det(B)

B. det(B)

C. B

D. det(A).B

Q.3 Let A be n x n matrix, then A is invertible if and only if

A. det A is zero

B. det A is not zero

Q.4 An n x n matrix A is diagonalizable if and only if A has ___ linearly independent eigenvectors.

A. n - 1

B. n

C. n + 1

D. n x n

Q.5 {\rm{If}}\,{\rm{AB = I = BA \,\ for }}\,\ matrices{\rm{ }}A,\,B\;and\,I,{\rm{ }}where\;\,I\,is\,\ an \,\ identity \,\ matrix, \,\ then

A. A is inverse of B,

B. B is inverse of A,

C. All of the above.

D. {A^{(- 1)}} = B,{B^{(- 1)}} = A,

Q.6 Let A be a real 2 by 2 matrix with complex eigen values \lambda=a-b+{i}, (b\neq 0) and associated eigenvectors v in C^2, then A=PCP^{-1}, where P=----------

A. P=[Rev -Imv]

B. P=[-Rev -Imv]

C. P=[Rev Imv]

D. P=[-Rev Imv]

Q.7 A null space is a vector space.

A. False

B. True

Q.8 Every square matrix A has at least one ----------.

A. Critical point

B. Fixed point

Q.9 We can add the matrices of _____.

A. same order

B. same number of rows

C. same number of columns

D. different order

Q.10 Let A be an n by n matrix whose distinct eigen values are \lambda_{1}, \lambda_{2},..., \lambda_{p}. For 1 \leq k \leq p, the dimension of the eigen space for \lambda_{k} is -------- than or ------- to the multiplicity of the eigen value \lambda_{k}

A. Less, equal

B. Greater, equal

Q.1 Every square matrix A has at least one ----------.

A. Fixed point

B. Critical point

Q.2 If u + v =u + w, then:

A. v + w.

B. v ≠ w

C. None of the above

D. v = w.

Q.3 A system of linear equations is said to be homogeneous if it can be written in the form _____.

A. AX=B

B. AX=0

Q.4 A matrix A is diagonalizable if and only if there are enough eigenvectors to form a basis of

A. R^3

B. R^n

C. R^2

D. None of the above

Q.5 Two equivalent vectors must have the same initial point:

A. True

B. False

C. May be.

D. None of the above

Q.6 Let A be the matrix of order 2X3 and B be the matrix of order 3X5,then which of the following is the order of the matrix AB?

A. 3X5

B. 3X3

C. 2X3

D. 2x5

Q.7 Let A be n x n matrix, then A is invertible if and only if

A. det A is zero

B. det A is not zero

Q.8 Let A be an n \times n matrix whose distinct eigen values are \lambda_1, \lambda_2, ..., \lambda_p. The matrix A is diagonalizable if and only if the ---------- of the dimensions of the distinct eigen spaces is equal to n

A. Product

B. sum

Q.9 If one of the Eigenvalues of {\left[A \right]_{n \times n}} is zero, it implies

A. The solution to [A][X]=[0]

B. The determinant of [A] is nonzero

C. The solution to [A][X]=[C] system of equations is unique

D. The determinant of [A] is zero

Q.10 The trace and determinant of a 2 × 2 matrix are known to be -2 and -35 respectively. Its Eigen values are

A. -7 and 5.

B. -30 and -5

C. 17.5 and -2

D. -37 and -1

Q.1 If x - 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

A. 2

B. 1/2

C. 0

D. -2

Q.2 If 3 is an eigenvalue of A and x is a corresponding eigenvector, then what is the eigenvalue of A^2 ?

A. 6

B. 12

C. 3

D. 9

Q.3 An n \cross n matrix with ------distinct eigenvalues is diagonalizable.

A. n

B. none of the above

C. n+1

D. n+2

Q.4 An n \times n matrix A is said to be diagonalizable if and only if A has n ---------- eigenvectors.

A. Linearly Independent

B. Linearly dependent

Q.5 The eigenvalues of a 4 by 4 matrix [A] are given as 2, -3, 1, 3 and 7. The |det(A)| then is

A. 25

B. 546

C. 530

D. cannot be determined

Q.6 Let A be a real 2 by 2 matrix with complex eigen values \lambda = a - bi(b \ne 0) and associated eigenvectors v in {C^2} , then

A. P = AC{P^{ - 1}}

B. A = {P^{ - 1}}CP

C. A = PC{P^{ - 1}}

D. None of the above

Q.7 The null-space of a linear transformation T is the set of all vectors v such that T (v) = _____.

A. negative

B. 0

C. non-zero

D. positive

Q.8 A matrix A is diagonalizable if and only if there are enough eigenvectors to form a basis of

A. R^3

B. R^2

C. R^n

D. None of the above

Q.9 If \lambda is an eigenvector of A, then every nonzero vector x such that Ax=\lambda x is called an ---------- of A corresponding to ----------

A. Eigenvector,

B. Eigenvalue, A

C. Eigenvector, A

D. Eigenvalue,

Q.10 If n x n matrices A and B are similar, then they have the _____ eigenvalues (with the same multiplicities).

A. multiplicative inverse of each other

B. distinct

C. additive inverse of each other

D. same

Q.1 Which one of the following statements is true for all real symmetric matrices?

A. All the eigenvalues are real.

B. All the eigenvalues are positive.

C. All the eigenvalues are distinct

D. Sum of all the eigenvalues is zero.

Q.2 If A and B are n x n matrices, then det (AB) = _____.

A. det(A).det(B)

B. det(B)

C. det(A).B

D. B

Q.3 If \lambda - 2 is a factor of the characteristic polynomial of matrix A , then which of the following is the Eigenvalue of A ?

A. {1}{2}

B. 0

C. 2

D. -2

Q.4 A matrix A is diagonalizable if and only if there are enough eigenvectors to form a basis of

A. R^2

B. R^3

C. None of the above

D. R^n

Q.5 An n x n matrix with n distinct eigen values is ____.

A. symmetric

B. diagonalizable

C. invertible

D. Hermitian

Q.6 Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a non-zero vector v such that Av = _____

A. x . v

B. x . A

C. x. Av

D. xv

Q.7 Two equivalent vectors must have the same initial point:

A. false

B. May be

C. None of the above

D. true

Q.8 A and its transpose matrix have _____ eigenvalues.

A. different

B. different and positive

C. same but opposite in sign

D. same

Q.9 If A= transpose of A(where A is a square matrix), then which of the following is the most appropriate option for A?

A. A is invertible matrix.

B. A is symetric matrix.

C. A is singular matrix.

D. A is a scalar matrix.

Q.10 Let A be an n by n matrix whose distinct eigen values are \lambda_{1}, \lambda_{2},..., \lambda_{p}. For 1 \leq k \leq p, the dimension of the eigen space for \lambda_{k} is -------- than or ------- to the multiplicity of the eigen value \lambda_{k}

A. Greater, equal

B. Less, equal

Q.1 If \lambda is an eigenvalue of a matrix A and x is a corresponding eigenvector, and if k is any positive integer, then {\lambda ^k} is an eigenvalue of _____ and x is a corresponding eigenvector.

A. A

B. ^k

C. [Image option]

D. A^k

Q.2 A column replacement operation on A does not change the _____.

A. row

B. determinant

C. matrix

D. column

Q.3 If n x n matrices A and B are similar, then they have the _____ characteristic polynomial.

A. inverse of the other

B. same

C. different

D. same but opposite in sign

Q.4 Let A be the matrix of order 2X3 and B be the matrix of order 3X5,then which of the following is the order of the matrix AB?

A. 3X5

B. 3X3

C. 2x5

D. 2X3

Q.5 The set of all solutions of (A-\lambdaI)x is just the ..... Of the matrix A

A. Topological space

B. Metric space

C. None of the above

D. Null space

Q.6 The eigenvalues of a 4 by 4 matrix [A] are given as 2, -3, 1, 3 and 7. The |det(A)| then is

A. cannot be determined

B. 25

C. 530

D. 546

Q.7 What is eigen value?

A. It is the inverse of the transform

B. A scalar associated with a given linear transformation

C. A matrix determined from the algebraic equations

D. A vector obtained from the coordinates

Q.8 What is Eigen value?

A. A vector obtained from the coordinates

B. A matrix determined from the algebraic equations

C. It is the inverse of the transform

D. A scalar associated with a given linear transformation

Q.9 A null space is a vector space.

A. True

B. False

Q.10 A row interchange _____ the sign of the determinant.

A. does not change

B. changes

Q.1 Two vectors are linearly dependent if and only if they lie_____

A. on a line parallel to x-axis

B. on the same line through origin

Q.2 We can add the matrices of _____.

A. same order

B. same number of columns

C. same number of rows

D. different order

Q.3 Every square matrix A has at least one ----------.

A. Critical point

B. Fixed point

Q.4 If A is triangular, then det A is the product of the entries on the _____ of A.

A. rows

B. main diagonal

C. columns

D. upper diagonal

Q.5 If \lambda is an eigenvector of A, then every nonzero vector x such that Ax=\lambda x is called an ---------- of A corresponding to ----------

A. Eigenvalue, A

B. Eigenvector, A

C. Eigenvalue,

D. Eigenvector,

Q.6 An n x n matrix A is diagonalizable if and only if A has ___ linearly independent eigenvectors.

A. n + 1

B. n - 1

C. n

D. n x n

Q.7 The determinant of a triangular matrix is the sum of the entries of the main diagonal.

A. True

B. False

Q.8 An n \times n matrix A is said to be diagonalizable if and only if A has n ---------- eigenvectors.

A. Linearly Independent

B. Linearly dependent

Q.9 A and its transpose matrix have _____ eigenvalues.

A. same but opposite in sign

B. different

C. different and positive

D. same

Q.10 If \lambda + 7 is a factor of the characteristic polynomial of a matrix C , then which of the following is the Eigenvalue of C ?

A. 7

B. {1}{7}

C. 0

D. -7

Q.1 If A is m\cross n and b is in R^n, a least square solution of Ax=b is an \widehat x in R^n such that

A. ||b-A x ||<=||b-Ax|| x R^n

B. ||b-A x||>=||b-Ax|| x R^n

Q.2 Each pair of eigenvalue and its corresponding eigenvectors provides a solution of the equation x' = Ax which is called

A. vector space

B. dynamical system

C. solution matrix

D. eigenfunctions

Q.3 The matrix M = \left[{\,\begin{array}{*{20}{c}} a&4&5&6\\\7&a&4&5\\\8&7&a&4\\\9&8&7&a \end{array}\,} \right] is a _____ matrix.

A. Diagonal

B. Toeplitz

C. Identity

D. Rectangular

Q.4 A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are _____.

A. Nonzero skew symmetric

B. Zero

C. Unit

D. Nonzero symmetric

Q.5 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is an orthonormal basis, then

A. o{j_w}y = (y.y){u_1} + (y.y){u_2} + ... + (y.y){u_p}

B. o{j_w}y = (y.{u_1}){u_1} + (y.{u_2}){u_2} + ... + (y.{u_p}){u_p}

Q.6 The product of upper triangular matrices is:

A. Lower triangular matrix

B. Diagonal matrix.

C. None of the above

D. Upper triangular matrix.

Q.7 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is any orthogonal basis of W, then \mathop y\limits^\^ = {c_1}{u_1} + {c_2}{u_2} + ... + {c_n}{u_n} Where

A. {c_j} = {{y.{u_j}}}{y}

B. None of these

C. {c_j} = {{y.{u_j}}}{{{u_j}.{u_j}}}

D. {c_j} = {{y.{u_j}}}{{u_j}}

Q.8 For u, v vectors in {R^n}, the distance between u and v is written as

A. dist(u,v) = ||u - u||

B. None of these

C. dist(u,v) = ||v-v||

D. dist(u,v) = ||u - v||

Q.9 Let W is a finite dimensional subspace of an inner product space V and y is any vector in V. The best approximation to y from W is then

A. Proj^{w}_{y}

B. Proj^{y}_{w}

Q.10 If there is a vector u=(1, -1, 2) then ||u|| is------

A. {7}

B. {6}

C. 2

D. 0

Q.1 The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.

A. singular

B. invertible

C. scalar

D. symmetric

Q.2 A vector whose length is 1 is called ----------

A. Identity Vector

B. Unit Vector

Q.3 Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation x'=Ax which is called ---------- of the differential equation.

A. Eigensolution

B. Eigenfunction

Q.4 Let V be an vector space, and let W be a subset of V. What does it mean when we say that W is closed under addition?

A. Whenever x and y are in V, then x+y is in V

B. W(x+y)=Wx+Wy for every two vectors x and y

Q.5 A square matrix with _____ columns is an orthogonal matrix.

A. same

B. orthogonal

C. even

D. orthonormal

Q.6 A _____ matrix with orthogonal columns is an orthogonal matrix.

A. diagonal

B. square

C. singular

D. rectangular

Q.7 A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are _____.

A. Nonzero skew symmetric

B. Nonzero symmetric

C. Unit

D. Zero

Q.8 If there is a vector v=(2, 1, 0) then ||v|| is ------

A. 0

B. {5}

C. 3

D. 2

Q.9 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is any orthogonal basis of W, then \mathop y\limits^\^ = {c_1}{u_1} + {c_2}{u_2} + ... + {c_n}{u_n} Where

A. {c_j} = {{y.{u_j}}}{{u_j}}

B. {c_j} = {{y.{u_j}}}{{{u_j}.{u_j}}}

C. {c_j} = {{y.{u_j}}}{y}

D. None of these

Q.10 The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are _____.

A. linearly dependent

B. linearly independent

Q.1 If a ____ matrix has orthonormal columns, then it also has orthonormal rows.

A. diagonal

B. square

C. rectangular

D. singular

Q.2 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is any orthogonal basis of W, then \mathop y\limits^\^ = {c_1}{u_1} + {c_2}{u_2} + ... + {c_n}{u_n} Where

A. {c_j} = {{y.{u_j}}}{{u_j}}

B. None of these

C. {c_j} = {{y.{u_j}}}{{{u_j}.{u_j}}}

D. {c_j} = {{y.{u_j}}}{y}

Q.3 Let W be a subspace of R^n, y be any vector in R^n and \widehat y the orthogonal projection of y onto W. Then \widehat y is the closest point in W to y, in the sense that

A. ||y- y||<||y-v||

B. ||y- y ||>||y-v||

Q.4 If \{u_{1}, u_{2}, ..., u_{p}\} is an orthonormal basis for a subspace W of R^n, then ...

A. Proj_{w}y=(y.u_{1})u_{p}+(y.u_{2})u_{p}+...+(y.u_{p})u_{p}

B. Proj_{w}y=(y.u_{1})u_{1}+(y.u_{2})u_{2}+...+(y.u_{p})u_{p}

Q.5 A matrix [A]_{n \times n has both positive and negative eigenvalues so in this case origin behaves as a ----------

A. Critical point

B. Saddle point

Q.6 If two rows are _____, they are linearly independent.

A. identical

B. orthonormal

C. orthogonal

D. perpendicular

Q.7 If u=(2, 0, -1) and v=(0, -1, 0) is an orthogonal set then which of the following has a norm of 1?

A. v.u

B. v.v

C. u.u

D. u.v

Q.8 For u, v vectors in {R^n}, the distance between u and v is written as

A. dist(u,v) = ||v-v||

B. dist(u,v) = ||u - u||

C. dist(u,v) = ||u - v||

D. None of these

Q.9 Which of the following set of vectors is an orthogonal set?

A. (1, 1, 1) ;(1, 0,-1)

B. (1, 0, 1) ;(-1, 0, -1)

C. (0, 1, 1) ;(1, 0,-1)

D. (1, 1, 0) ;(1, 0,-1)

Q.10 Let V be a one Eigen vector then conjugate eigen vector is represented by,

A. V^{__}

B. V^

C. V

D. V'

Q.1 Elementary row operations on a matrix do not affect the -------- relations among the columns of the matrix.

A. Linear independence

B. Linear dependence

Q.2 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is an orthonormal basis, then

A. o{j_w}y = (y.y){u_1} + (y.y){u_2} + ... + (y.y){u_p}

B. o{j_w}y = (y.{u_1}){u_1} + (y.{u_2}){u_2} + ... + (y.{u_p}){u_p}

Q.3 Suppose that real solutions y_1 and y_2 of x'=Ax, form a basis for the two-dimensional real vector space if y_1 and y_2 are .....

A. Linearly Independent

B. Linearly dependent

Q.4 If there is a vector u=(1, -1, 2) then ||u|| is------

A. 0

B. 2

C. {7}

D. {6}

Q.5 Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation x'=Ax which is called ---------- of the differential equation.

A. Eigen vectors

B. Eigen functions

Q.6 Let V be an vector space, and let W be a subset of V. What does it mean when we say that W is closed under addition?

A. W(x+y)=Wx+Wy for every two vectors x and y

B. Whenever x and y are in V, then x+y is in V

Q.7 The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are linearly independent.

A. TRUE

B. FALSE

Q.8 Which of the following is true for the matrix M = \left[\begin{array}{l} \,{M_{\,11}}\,\,\,\,\,\,\,\,\,\,\,{M_{\,12}}\,\,\,\,\,\,\,\,\,\,\,{M_{\,13}}\\\ O\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,{M_{\,22}}\,\,\,\,\,\,\,\,\,\,\,{M_{\,23}}\\\ O\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,O\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,\,{M_{\,33}}\, \end{array} \right]\,; where {M_{\,\,11}} , {M_{\,22}} and {M_{\,33}} are square sub-matrices , and O is a zero sub-matrix ?

A. It is diagonal-constant matrix.

B. It is a block upper triangular matrix.

C. It is a block lower triangular matrix.

D. It is a Null matrix.

Q.9 If the columns of a matrix are _____ then the matrix is invertible.

A. linearly independent

B. linearly dependent

Q.1 Which of the following set of vectors is an orthogonal set?

A. (1, 1, 0) ;(1, 0,-1)

B. (1, 0, 1) ;(-1, 0, -1)

C. (0, 1, 1) ;(1, 0,-1)

D. (1, 1, 1) ;(1, 0,-1)

Q.2 A m ×n matrix U has orthonormal columns if and only if {U^t}U = I

A. False

B. True

Q.3 The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.

A. invertible

B. singular

C. symmetric

D. scalar

Q.4 If x is orthogonal to both u and v, then x must be _____ to u – v.

A. orthogonal

B. equal

C. orthonormal

D. parallel

Q.5 Elementary row operations on a matrix do not affect the -------- relations among the columns of the matrix.

A. Linear independence

B. Linear dependence

Q.6 If u=(1, k, 0) and v=(1, 1, 0) are orthogonal vectors then which of the following is the value of k?

A. k = 2

B. k = -1

C. k = 1

D. k = 0

Q.7 Let W be a subspace of {R^n} and \{ {u_1},{u_2},...,{u_p}\} is any orthogonal basis of W, then \mathop y\limits^\^ = {c_1}{u_1} + {c_2}{u_2} + ... + {c_n}{u_n} Where

A. None of these

B. {c_j} = {{y.{u_j}}}{{{u_j}.{u_j}}}

C. {c_j} = {{y.{u_j}}}{{u_j}}

D. {c_j} = {{y.{u_j}}}{y}

Q.8 Which of the following set of vectors is NOT an orthogonal set?

A. (2, 3) ;(-6, 4)

B. (1, 2, 0) ;(1, 0, 3)

C. (3, 0, 0) ;(0, 1, 0)

D. (1, 1, 1) ;(1, 0,-1)

Q.9 The zero vector is orthogonal to every vector in R^n

A. True

B. False

Q.10 If s = \{ {u_1} + {u_2} + ...{u_p}\} is an orthogonal basis for a subspace W of {R^n} Then each y in W can be uniquely expressed as a linear combination of {u_1} + {u_2} + ...{u_p} That is{c_1}{u_1} + {c_2}{u_2} + ...{c_p}{u_p} Where\frac{{y.{u_j}}}{{{u_j}.{u_j}}}

A. False

B. True
